

Algebra 1: Polynomials

Homework₁ *Level*

Name: _____

Date: _____

Question 1: Solve for x : $x^2 + 5x + 6 = 0$

- A) $x = -2, x = -3$
- B) $x = -1, x = -6$
- C) $x = 2, x = 3$
- D) $x = 1, x = 6$
- E) $x = -2, x = 3$

Question 2: Factor the polynomial: $x^2 + 7x + 12$

- A) $(x + 3)(x + 4)$
- B) $(x + 2)(x + 6)$
- C) $(x + 1)(x + 12)$
- D) $(x - 3)(x - 4)$
- E) $(x + 3)(x - 4)$

Question 3: Simplify the expression: $(2x^2 + 3x - 1) + (x^2 - 2x - 1)$

- A) $3x^2 + x - 2$
- B) $2x^2 + x - 1$
- C) $3x^2 + 5x - 2$
- D) $x^2 + 5x - 2$
- E) $x^2 - x - 2$

Question 4: Find the roots of the equation: $x^2 - 4x - 5 = 0$ using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = -4$, and $c = -5$

- A) $x = -1, x = 5$
- B) $x = 1, x = -5$
- C) $x = 2, x = -2$
- D) $x = -2, x = 2$
- E) $x = -3, x = 3$

Question 5: Solve for y : $y^2 - 9y + 20 = 0$

- A) $y = 4, y = 5$
- B) $y = 1, y = 9$
- C) $y = 2, y = 10$
- D) $y = 5, y = 4$

E) $y = 2, y = 8$

Question 6: Factor the polynomial: $x^3 + 2x^2 - 7x - 12$

- A) $(x + 1)(x + 3)(x - 4)$
- B) $(x + 2)(x - 3)(x + 4)$
- C) $(x + 3)(x - 2)(x + 2)$
- D) $(x - 1)(x - 3)(x + 4)$
- E) $(x + 1)(x - 3)(x + 4)$

Question 7: Simplify the expression: $(x^2 + 2x - 1)(x + 3)$ using the distributive property

- A) $x^3 + 5x^2 + 5x - 3$
- B) $x^3 + 5x^2 - x - 3$
- C) $x^3 + 2x^2 - x - 3$
- D) $x^3 + x^2 - x - 3$
- E) $x^3 + 5x^2 + x - 3$

Question 8: Find the roots of the equation: $x^2 + 2x - 6 = 0$ using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = 2$, and $c = -6$

- A) $x = -1 - \sqrt{7}, x = -1 + \sqrt{7}$
- B) $x = -2 - \sqrt{2}, x = -2 + \sqrt{2}$
- C) $x = 2 - \sqrt{2}, x = 2 + \sqrt{2}$
- D) $x = 1 - \sqrt{3}, x = 1 + \sqrt{3}$
- E) $x = -3 - \sqrt{5}, x = -3 + \sqrt{5}$

Question 9: Factor the polynomial: $x^4 - 2x^3 - 13x^2 + 14x + 12$

- A) $(x - 2)(x + 1)(x + 2)(x - 3)$
- B) $(x + 2)(x - 1)(x + 3)(x - 2)$
- C) $(x - 1)(x + 2)(x + 1)(x - 3)$
- D) $(x + 1)(x - 3)(x + 2)(x + 2)$
- E) $(x + 1)(x - 2)(x + 2)(x - 3)$

Question 10: Prove that $(x + y)^2 = x^2 + 2xy + y^2$ using algebraic manipulation and explain the concept of binomial expansion. Show all the steps and provide explanations for each step.

Chef's Tips (Hints):

- To factor a quadratic expression, look for two numbers whose product is the constant term and whose sum is the coefficient of the linear term.
- The quadratic formula can be used to solve quadratic equations of the form $ax^2 + bx + c = 0$, where a , b , and c are constants.
- When simplifying expressions, combine like terms and use the distributive property to eliminate parentheses.